

Bernstein decomposition & centre

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Notation:

- $G = G(F)$ p -adic reductive; F p -adic field
- $\text{Rep}(G)$ cat. of smooth reps of G / $\mathbb{C}$
- $\mathcal{H}$ Hecke algebra of G (cptly supp. locally const. func.)

Recall $\text{Rep}(G) \simeq$ cat. of non-degenerate left $\mathcal{H}$ -mod

$$P = LU \quad , \quad \iota_P^G: \text{Rep}(L) \rightarrow \text{Rep}(G)$$

$$- \quad J_U: \text{Rep}(G) \rightarrow \text{Rep}(L)$$

- $V \in \text{Rep}(G)$ is supercuspidal (cuspidal?) if $J_U(V) = 0$ for all proper parabolics P .

Let $\text{Rep}^{\text{cusp}}(G)$ be the cat. of cusp reps.

Thm. Let $\pi \in \text{Rep}(G)$ be irr.

- 1) $\exists$ a Levi L of G and an irr. cusp. rep. σ of L s.t. π is isom. to a subqt of $\iota_P^G \sigma$.
- 2) The pair (L, σ) assoc. to π as in (1) is unique up to conjugacy. i.e. if there is another (M, τ) , then $\exists g \in G$ s.t. $M = g^L L g$ and $\tau = \sigma^g$.

Defn The equiv. class of (L, σ) is called the cuspidal support of π .

pf Choose a parabolic $P = LU$ of G minimal wr.t. $J_U(\pi) \neq 0$.

The Jacquet functor is transitive, in the sense that if $M \subset L$ contained in $\text{Lev of } L$

parabolic $Q = MN$ of G , then $J_N = J_{L \cap N} \circ J_U$

By transitivity of J and minimality of P , $J_U(\pi)$ is a (usp. rep. of L .

J_U sends fin. gen. reps to fin. gen. reps, so $J_U(\pi)$ has some irred. cuspidal quotient

σ - By adjunction, get $\pi \hookrightarrow \mathcal{L}_P^G(\sigma)$

(2) Suppose (M, τ) is another w/ this property. so π is a subqt of $\mathcal{L}_Q^G \tau$.

$$Q = MN$$

J_U is exact, so $J_U(\pi)$ is a subqt of $J_U \mathcal{L}_Q^G(\tau)$

Prop (Mackey decomp. for (usp. reps)) Suppose (M, τ) is (usp, $J_U \mathcal{L}_Q^G$ has a nonzero (usp. component. Then $(J_U \mathcal{L}_Q^G)(\tau)$ has Jordan-Hölder decomposition w/ factors $\bigoplus_{g \in G/M} g\tau$ assoc. graded

$$L \backslash \{g \in G: g^{-1} M g = L\} / M$$

$\Rightarrow$ (2) σ is also a subqt of $J_U \mathcal{L}_Q^G \tau$, so it is one of the $g\tau$. $\square$

We enlarge the class (L, σ) by saying that $(L, \sigma) \sim (L, \sigma \otimes \chi)$ for

χ unramified. $\chi: G \rightarrow \mathbb{C}^\times$ is trivial on all open cpt subgps.

$$\text{Let } G^0 = \{g \in G: \text{val}_F(\chi(g)) = 0, \forall \chi \in \chi_F(G)\}$$

Facts: G^0 contains all cpt opens of G

G/G^0 is free abelian of finite rank.

ZG^0 has finite index in G

$$X_{\text{ur}}(G) = \text{Hom}(G/G^0, \mathbb{C}^\times)$$

alg. torus

Fix a smooth irrep $\pi \in \text{Rep } G$, we get an orbit (L, σ) under twisting by

$$X_{\text{ur}}(G). \quad [L, \sigma]_G.$$

If $\nu \in X_{\text{ur}}(G)$, $\nu|_Z = 1$, then $\nu|_Z = 1 \Rightarrow \text{Stab}_{X_{\text{ur}}(G)}(\nu)$ is finite

So orbit $[L, \sigma]_G$ may be viewed as an algebraic variety by quotienting $X_{\text{ur}}(G)$ by stabilizer + conjugation

$[L, \sigma]_G$ is an "inertial support".

Let B be the set of inertial supports.

For $s \in B$, let Rep_s^G be the full subcat. of objects π for which all ir. subqts have inertial support s .

Thm (Bernstein decomposition)

$$\text{Rep } G = \prod_{s \in B} \text{Rep}_s^G(G) \quad \text{and the factors are indecomposable}$$

Prop. $\text{Rep } G = \text{Rep}^{\text{cusp}} G \times \text{Rep}^{\text{nc}} G$
 $\uparrow$
 full subcat. of objects
 $\hookrightarrow$ no cusp. subcat.

Moreover, $\text{Rep}^{\text{cusp}}(G) = \prod_{s=[\alpha, \pi] \in B} \text{Rep}^s(G).$

Pt Bernstein decomp given above:

Let $V \in \text{Rep}(G)$. If V is of the form $V = \bigoplus_S V^S$, $V^S \in \text{Rep}^S(G)$,
 say V is split. We will prove all reps are split by embedding V into
 a split rep and then noting that subreps of split are split.

Fix a minimal parabolic $P_{\min} = L_{\min} U_{\min}$ of G . Only consider
 Levi's / parabolics that are standard w.r.t. $L_{\min} / P_{\min}$, i.e. $P \supset P_{\min}$ and
 L is the unique Levi of P containing $L_{\min}$.

Let $\mathcal{C}(G) = \prod_{L \text{ std}} \text{Rep}^{\text{cusp}}(L)$

Componentwise induction $\iota: \mathcal{C}(G) \rightarrow \text{Rep}(G)$

$$\iota((W_L)) = \bigoplus_{P \text{ std}} \iota_P^G(W_L)$$

Similarly get $J: \text{Rep } G \rightarrow \mathcal{C}(G)$, $J(v) = (J_U(v))_P^{\text{cusp}}$

Claim (J, ι) are an adjoint pair:

$$\text{Hom}_{(G)}(J(V), W) = \prod_L \text{Hom}_L(J_u(V)^{\text{cusp}}, W_L)$$

(because W_L is cuspidal already) $= \prod_L \text{Hom}_L(J_u(V), W_L)$

adjunction $= \prod_L \text{Hom}_G(V, \iota_P^G(W_L))$

There are only finitely many standard parabolics.

$$= \text{Hom}_G(V, \iota(W))$$

ι, J are exact

J is faithful: if $V \neq 0$, then $\exists$ minimal unipotent among standard parabolics

s.t. $J_u(V) \neq 0 \Rightarrow J_u(V)$ is cuspidal. $\Rightarrow J(V) \neq 0$.

Claim unit of adjunction $\eta_V: V \rightarrow (\iota \circ J)(V)$ is injective.

Suppose $\ker(\eta_V) = K$.

$$\begin{array}{ccc} V & \xrightarrow{\eta} & (\iota \circ J)(V) \\ \uparrow & \lrcorner & \uparrow \\ K & \xrightarrow[\eta|_K=0]{} & (\iota \circ J)(K) \end{array}$$

$$\begin{array}{ccc} \eta_K \in \text{Hom}(K, (\iota \circ J)(K)) & \xrightarrow{\sim} & \text{Hom}(J(K), J(K)) \\ \parallel & & \\ 0 & \xrightarrow{\quad \quad \quad} & \text{id} = 0 \end{array}$$

$$\Rightarrow J(K) = 0 \Rightarrow K = 0.$$

Hence $V \hookrightarrow (\iota \circ J)(V)$. Want $(\iota \circ J)(V)$ is split. By cuspidal decomp.,

$$J_u(V)^{\text{cusp}} = \bigoplus_{\substack{\text{irreducible classes} \\ \text{of reps of } L}} W_t, \quad W_t \in \text{Rep}^t(L)$$

$$\rightarrow (L \circ J)(V) = \bigoplus_{L \text{ std}} L_P^G \left(\bigoplus_t W_t \right)$$

However, we can pass the induction past the inner $\bigoplus$.

$$(L \circ J)(V) = \bigoplus_{L \text{ std}} \bigoplus_t L_P^G(W_t)$$

Prop. Let $t = [L, \sigma]_L$ be an inertial pair for L

$s = [L, \sigma]_G$ considered as an inertial pair for G

Then L_P^G sends $\text{Rep}^t(L) \rightarrow \text{Rep}^s(G)$.

This implies $(L \circ J)(V)$ splits.

Proof. Suppose π is an irred. subgt of $L_P^G(W)$ for some $W \in \text{Rep}^t(L)$.

(Cuspidal supports: $\exists$ parabolic $Q = MN$, ir. (asp. rep τ of M s.t. $\pi \hookrightarrow L_Q^G \tau$).

By adjunction, τ is an irred. gt of $J_N \pi$. Exactness of J : $J_N \pi$ is isom. to a subgt of $J_N L_P^G(W)$, so τ is also an irred. subgt of this.

Mackey decomp. $\Rightarrow g(J_N L_P^G(W)) = \bigoplus_{g \in M \backslash \{g \in G: g^{-1}Lg = M\}/L} gW$

Hence τ is an irred. subgt of one of these gW .

$\Rightarrow$ inertial type of π is $[L, \tau]_G$.

Finally, pt that subreps of a split rep are split, is elementary.

Recall the centre of a cat. is the ring of endomorphisms of the identity functor.

A product category decomposition as above induces a product decomposition of its centre.

Let $\mathcal{H}$ = Hecke alg For each open cpt $K \subset G$, idempotent e_K

Define a partial ordering on idempotents of $\mathcal{H}$ by $e \leq f \Leftrightarrow e\mathcal{H}e \subset f\mathcal{H}f$

Topologise $\mathcal{H}$ by letting $\mathcal{H}(1-e)$ be a system of nbhd's of 0, then form the completion $\hat{\mathcal{H}}$ w.r.t. this topology.

Since $\mathcal{H} = \mathcal{H}e \oplus \mathcal{H}(1-e)$, can also define $\hat{\mathcal{H}} = \varprojlim_e \mathcal{H}e$, $\mathcal{H}f \rightarrow \mathcal{H}e$

$\hat{\mathcal{H}}$ has a unit $(e)_e$ $x \mapsto xe$

Lemma $\hat{\mathcal{H}}$ is the ring of endomorphism of the forgetful functor from $\text{Rep}(\mathcal{H})$ to vector spaces.

Sketch. $\text{Rep}(\mathcal{H}) \simeq$ nondeg left $\mathcal{H}$ -mod, and this has a nat'l $\hat{\mathcal{H}}$ -action.
 $\uparrow$
 $\mathcal{H}$ generates

So we prove the claim for $\text{End}_{\text{v.s.}}(\mathcal{H})$.

$\hat{\mathcal{H}}$ embeds into the set of endomorphisms that commute w/ right multiplication by $\mathcal{H}$.

For all $\varphi: \mathcal{H} \rightarrow \mathcal{H}$ as $\mathcal{H}$ -mod

$$\begin{array}{ccc} \text{Frg}(\mathcal{H}) & \xrightarrow{\varphi} & \text{Frg}(\mathcal{H}) \\ \hat{\mathcal{H}} \downarrow & & \downarrow \mathcal{H} \\ \text{Frg}(\mathcal{H}) & \xrightarrow{\varphi} & \text{Frg}(\mathcal{H}) \end{array}$$

However, given any φ w/ this property, can be realized by the action of $\hat{h} \in \hat{\mathcal{H}}$ defined by $\hat{h}e = \varphi(e)$. $\square$

Cor. $Z(\hat{H})$ is the centre of $\text{Rep}(H)$.

= inverse limit of $Z(eHe)$

Theorem $Z(\hat{H}) = \prod_{S \in B} Z(\hat{H})_S$

$Z(\hat{H})_S$ is isom. to the coord. ring of the variety assoc. to S .

Given $\hat{h} \in Z(\hat{H})_S$, get a function on variety by sending an irred. cusp.

rep π of H to the scalar by which $\hat{h}$ acts on π .

↳ cusp supp in S